NEET UG HIGH-YIELD COMPREHENSIVE TUTORIAL

Amines and Diazonium Salts

Complete NCERT-Aligned Theory, Structural Mechanics, Name Reactions, and NEET-Pattern Practice Assessment

1. Introduction and Biological Importance

Amines constitute a fundamental class of organic nitrogenous compounds formally regarded as derivatives of ammonia ($\mathrm{NH_3}$) wherein one, two, or all three hydrogen atoms are replaced by alkyl or aryl groups. In nature, amino moieties occur ubiquitously in proteins, vitamins, alkaloids, and hormones.

High-Yield NCERT Medical & Commercial Applications (Frequently Tested in NEET)

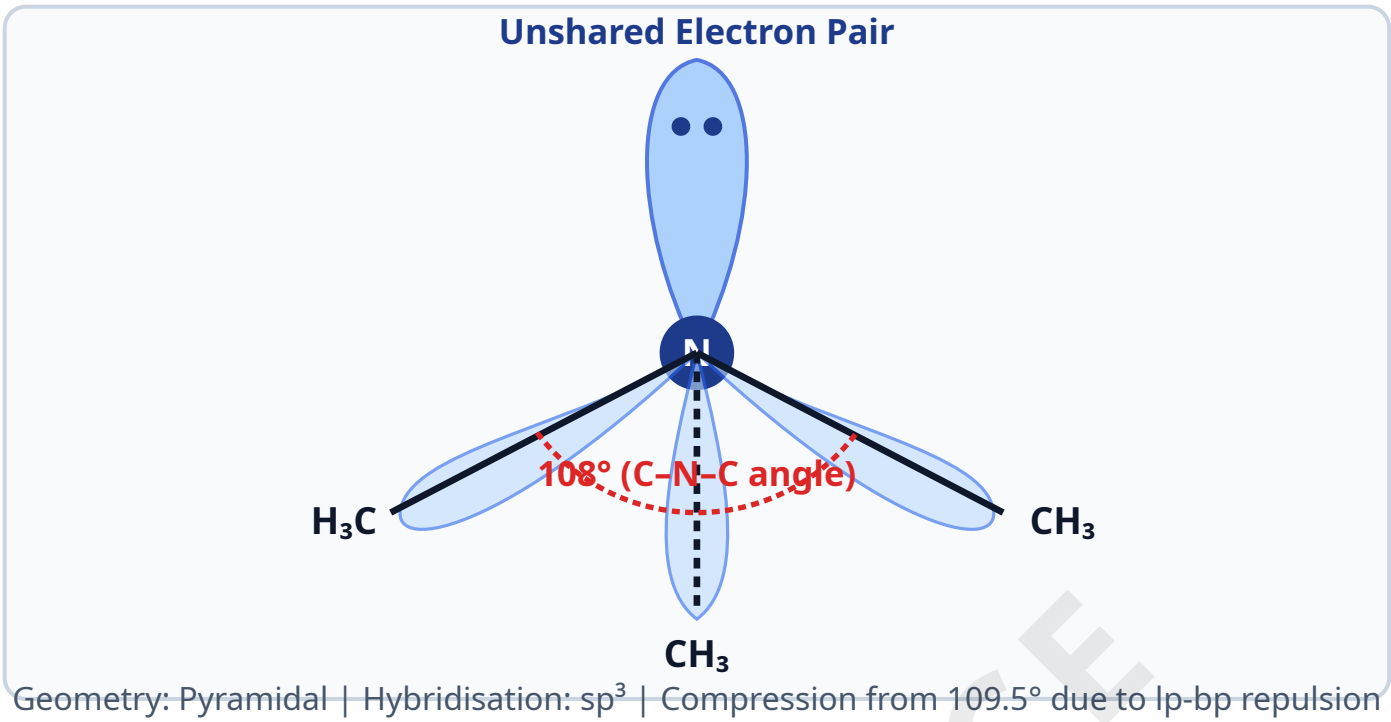
- **Adrenaline and Ephedrine:** Both compounds contain a *secondary amino group* (2°) and are clinically utilized to constrict blood vessels and increase blood pressure.
- **Novocain (Procaine):** A synthetic amino ester widely administered as a local anaesthetic in dentistry.
- **Benadryl (Diphenhydramine):** A ubiquitous antihistaminic pharmaceutical agent containing a *tertiary amino group* (3°).
- **Quaternary Ammonium Salts:** Long-chain cationic surfactants used as germicides and conditioning detergents.
- **Diazonium Salts:** Essential synthetic intermediates for creating azo dyes and functionalized aromatic rings.

2. Electronic Structure and Molecular Geometry

In amines, the central nitrogen atom is trivalent and carries one unshared pair of electrons (lone pair). The nitrogen atom adopts sp^3 **hybridisation**. The three sp^3 hybrid orbitals overlap with either the $1s$ orbital of hydrogen or sp^3 hybrid orbitals of carbon, while the fourth sp^3 orbital houses the lone pair.

Consequently, the geometry of amines is **pyramidal** (derived from a tetrahedral electron-pair geometry). Due to repulsive lone pair–bond pair interactions, which exceed bond pair–bond pair repulsion, the $\mathrm{C-N-E}$ bond angle (where $\mathrm{E = C}$ or H) is compressed to less than the ideal tetrahedral angle of 109.5°. In trimethylamine, $(\mathrm{CH_3})_3\mathrm{N}$, the $\mathrm{C-N-C}$ bond angle is precisely 108°.

BOSONANCE



Tutorial Diagram 1: Pyramidal molecular geometry and orbital arrangement of trimethylamine.

3. Classification and Systematic IUPAC Nomenclature

Amines are classified according to the number of carbon groups attached directly to the nitrogen atom:

- **Primary (1°) Amine:** Nitrogen bound to one alkyl or aryl group ($\mathrm{RNH_2}$ or $\mathrm{ArNH_2}$).
- **Secondary (2°) Amine:** Nitrogen bound to two carbon groups ($\mathrm{R_2NH}$ or $\mathrm{R-NH-R'}$).
- **Tertiary (3°) Amine:** Nitrogen bound to three carbon groups ($\mathrm{R_3N}$ or $\mathrm{R_2NR'}$ or $\mathrm{RNR'R''}$).
- **Simple vs. Mixed:** An amine is termed *simple* if all attached R groups are identical (e.g., diethylamine), and *mixed* if they differ (e.g., ethylmethylamine).

Structural Formula	Common Name	IUPAC Name	Class
$\mathrm{CH_3-CH_2-NH_2}$	Ethylamine	Ethanamine	1°
$\mathrm{CH_3-CH_2-CH_2-NH_2}$	*n*-Propylamine	Propan-1-amine	1°
$\mathrm{CH_3-CH(NH_2)-CH_3}$	Isopropylamine	Propan-2-amine	1°
$\mathrm{CH_3-NH-CH_2-CH_3}$	Ethylmethylamine	N-Methylethanamine	2°
$\mathrm{(CH_3)_3N}$	Trimethylamine	N,N-Dimethylmethanamine	3°
$\mathrm{C_2H_5-N(C_2H_5)-CH_2CH_2CH_2CH_3}$	N,N-Diethylbutylamine	N,N-Diethylbutan-1-amine	3°
$\mathrm{CH_2=CH-CH_2-NH_2}$	Allylamine	Prop-2-en-1-amine	1°
$\mathrm{H_2N-(CH_2)_6-NH_2}$	Hexamethylenediamine	Hexane-1,6-diamine	1° (diamine)
$\mathrm{C_6H_5-NH_2}$	Aniline	Aniline or Benzenamine	1° aryl
$\mathrm{2-(CH_3)C_6H_4NH_2}$	*o*-Toluidine	2-Methylbenzenamine	1° aryl
$\mathrm{4-Br-C_6H_4NH_2}$	*p*-Bromoaniline	4-Bromobenzenamine	1° aryl
$\mathrm{C_6H_5-N(CH_3)_2}$	N,N-Dimethylaniline	N,N-Dimethylbenzenamine	3° aryl

4. Methods of Preparation of Amines

1. Reduction of Nitro Compounds

Nitroalkanes and nitroarenes are smoothly converted to primary amines by catalytic hydrogenation using $\mathrm{H_2}$ over finely divided Ni, Pt, or Pd, or by reduction with active metals in acidic media:

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$$\mathrm{Ar-NO_2 + 3H_2 \xrightarrow{Pd/Ethanol} Ar-NH_2 + 2H_2O}$$
$$\mathrm{Ar-NO_2 \xrightarrow{Sn+HCl \quad or \quad Fe+HCl} Ar-NH_2}$$

NCERT Key Fact for NEET: Reduction with Fe **scrap and** HCl is preferred industrially over $\mathrm{Sn/HCl}$ because the $\mathrm{FeCl_2}$ formed undergoes hydrolysis during the course of the reaction to regenerate hydrochloric acid:

$$\mathrm{FeCl_2 + 2H_2O \longrightarrow Fe(OH)_2 + 2HCl}$$

Therefore, only a catalytic or small initiating amount of hydrochloric acid is required.

2. Ammonolysis of Alkyl Halides (Hofmann's Ammonolysis)

Cleavage of the carbon–halogen ($\mathrm{C-X}$) bond by ammonia via nucleophilic substitution ($\mathrm{S_N2}$) at $373\ \mathrm{K}$ in a sealed tube yields a primary amine:

$$\mathrm{NH_3 + R-X \longrightarrow R-\overset{+}{N}H_3\bar{X} \xrightarrow{NaOH} R-NH_2 + H_2O + NaX}$$

Because the resulting primary amine is itself an active nucleophile, it reacts further with unreacted alkyl halide to produce secondary and tertiary amines, ultimately culminating in quaternary ammonium salts:

$$\mathrm{RNH_2 \xrightarrow{RX} R_2NH \xrightarrow{RX} R_3N \xrightarrow{RX} R_4\overset{+}{N}\bar{X}}$$

- **To obtain 1° amine as the major product:** Use a *large excess* of ammonia.
- **Order of reactivity of alkyl halides:** $\mathrm{R-I > R-Br > R-Cl}$ (governed by $\mathrm{C-X}$ bond dissociation enthalpy).

3. Reduction of Nitriles (Ascent of Series)

Alkyl cyanides undergo reduction with lithium aluminium hydride ($\mathrm{LiAlH_4}$), catalytic hydrogenation ($\mathrm{H_2/Ni}$), or sodium amalgam with ethanol ($\mathrm{Na(Hg)/C_2H_5OH}$, known as Mendius reaction) to yield primary amines:

$$\mathrm{R-C\equiv N \xrightarrow{LiAlH_4 \quad or \quad H_2/Ni} R-CH_2-NH_2}$$

Synthetic utility: Adds exactly one carbon to the starting alkyl carbon chain (step-up reaction). For example, converting chloroethane to propan-1-amine via propanenitrile.

4. Reduction of Amides

Amides undergo clean reduction with lithium aluminium hydride to form primary amines retaining the identical number of carbon atoms:

$$\mathrm{R-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow[(ii)\ H_2O]{(i)\ LiAlH_4} R-CH_2-NH_2}$$

5. Gabriel Phthalimide Synthesis

A premier method for preparing pure **aliphatic primary amines** without contamination from secondary or tertiary amines.

1. Phthalimide is treated with ethanolic KOH to generate potassium phthalimide.
2. The phthalimide anion acts as a nucleophile, reacting with an alkyl halide ($\mathrm{R-X}$) via an $\mathrm{S_N2}$ pathway to yield N-alkylphthalimide.

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3. Alkaline hydrolysis of N-alkylphthalimide yields the corresponding pure primary amine and sodium phthalate.

$$\mathrm{C_6H_4(CO)_2NH} \xrightarrow{\mathrm{KOH}} \mathrm{C_6H_4(CO)_2N^-K^+} \xrightarrow{\mathrm{R-X}} \mathrm{C_6H_4(CO)_2N-R} \xrightarrow{\mathrm{NaOH(aq)}} \mathrm{R-NH_2} + \mathrm{C_6H_4(C}$$

Vital NEET Assertion-Reason Point: *Aromatic primary amines (such as aniline) cannot be prepared by Gabriel phthalimide synthesis.* This is because aryl halides ($\mathrm{Ar-X}$) possess partial double-bond character due to resonance and do not undergo nucleophilic aromatic substitution with the bulky phthalimide anion under standard conditions.

6. Hoffmann Bromamide Degradation Reaction

Treatment of an aliphatic or aromatic primary amide with bromine in an aqueous or ethanolic solution of sodium hydroxide causes degradation to a primary amine:

$$\mathrm{R-\overset{\overset{O}{\|}}{C}-NH_2} + \mathrm{Br_2} + 4\mathrm{NaOH} \longrightarrow \mathrm{R-NH_2} + \mathrm{Na_2CO_3} + 2\mathrm{NaBr} + 2\mathrm{H_2O}$$

- **Descent of series:** The amine formed contains **one carbon atom less** than the starting amide.
- **Mechanism detail:** Involves an intramolecular rearrangement in which the alkyl or aryl group migrates from the carbonyl carbon to the electron-deficient nitrogen atom (via acyl nitrene / isocyanate intermediate $\mathrm{R-N=C=O}$).
- **Stoichiometry for NEET:** Exactly 1 mole of $\mathrm{Br_2}$ and 4 moles of NaOH are consumed per mole of primary amide.

5. Physical Properties of Amines

Physical State, Colour, and Odour

- Lower aliphatic amines (methanamine, ethanamine) are colourless gases possessing a characteristic unpleasant "fishy" odour.
- Primary amines with three or more carbons are liquids; higher homologs containing more than eight carbon atoms are solids.
- Pure arylamines are generally colourless liquids or solids, but rapidly turn dark brown upon storage due to ease of atmospheric aerial oxidation.

Solubility in Water

Lower aliphatic amines dissolve readily in water due to their capacity to form intermolecular hydrogen bonds with polar water molecules ($\mathrm{R-NH_2\cdots H-O-H}$).

- **Chain length effect:** Solubility decreases sharply with increasing molecular mass because of the expanding non-polar, hydrophobic hydrocarbon chain. Higher amines are essentially insoluble in water.
- **Comparison with alcohols:** Alcohols are significantly *more soluble* in water than amines of comparable molecular mass. Oxygen is more electronegative (3.5) than nitrogen (3.0), creating a more polarized $\mathrm{O-H}$ bond that forms stronger hydrogen bonds with water compared to the $\mathrm{N-H}$ bond.

Boiling Points and Intermolecular Association

Primary and secondary amines engage in intermolecular hydrogen bonding via their polar $\mathrm{N-H}$ bonds.

- Primary amines have two hydrogen atoms available on nitrogen, resulting in extensive intermolecular association.
- Secondary amines possess only one active hydrogen atom, exhibiting moderate association.

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- Tertiary amines possess no hydrogen atom bound to nitrogen; consequently, they cannot undergo intermolecular hydrogen bonding with each other.

Order of Boiling Points for Isomeric Amines:
$$\text{Primary } (1^\circ) > \text{Secondary } (2^\circ) > \text{Tertiary } (3^\circ)$$

Compound	Type	Molar Mass ($\mathrm{g\,mol^{-1}}$)	Boiling Point (K)
$\mathrm{n-C_4H_9NH_2}$ (Butan-1-amine)	1° Amine	73	350.8
$\mathrm{(C_2H_5)_2NH}$ (Diethylamine)	2° Amine	73	329.3
$\mathrm{C_2H_5N(CH_3)_2}$ (Ethyldimethylamine)	3° Amine	73	310.5
$\mathrm{n-C_4H_9OH}$ (Butan-1-ol)	Alcohol	74	390.3
$\mathrm{C_2H_5CH(CH_3)_2}$ (2-Methylbutane)	Alkane	72	300.8

6. Chemical Reactions and Basicity Dynamics

The Fundamental Basicity of Amines

Amines behave as Lewis bases due to the unshared electron pair on the nitrogen atom. They react with aqueous mineral acids to produce water-soluble ionic ammonium salts:

$$\mathrm{R-NH_2 + HX \rightleftharpoons R-\overset{+}{N}H_3\bar{X}}$$

Treatment of these salts with a strong base (NaOH) regenerates the original free organic amine:

$$\mathrm{R-\overset{+}{N}H_3\bar{X} + \bar{O}H \longrightarrow R-NH_2 + H_2O + \bar{X}}$$

This salt-formation behavior is commonly used to separate basic amines from water-insoluble, non-basic organic compounds.

Quantitative Basicity (K_b and $\mathrm{p}K_b$)

In aqueous equilibrium:

$$\mathrm{R-NH_2 + H_2O \rightleftharpoons R-\overset{+}{N}H_3 + \bar{O}H}$$

$$K_b = \frac{[\mathrm{R-\overset{+}{N}H_3}][\mathrm{\bar{O}H}]}{[\mathrm{R-NH_2}]}, \qquad \mathrm{p}K_b = -\log_{10} K_b$$

Core Relationship: A larger K_b value or a **smaller** $\mathrm{p}K_b$ **value** designates a **stronger base**.

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Basicity Trends: Inductive Effect vs. Solvation vs. Steric Factors

1. In the Gas Phase (Isolated Molecules):

In the absence of a solvent, basicity is governed entirely by the electron-releasing inductive effect ($+\mathrm{I}$) of alkyl groups, which disperses the positive charge on the conjugate ammonium cation:

$$3^\circ \text{ Amine} > 2^\circ \text{ Amine} > 1^\circ \text{ Amine} > \mathrm{NH_3}$$
$$(\mathrm{C_2H_5})_3\mathrm{N} > (\mathrm{C_2H_5})_2\mathrm{NH} > \mathrm{C_2H_5NH_2} > \mathrm{NH_3}$$

2. In Aqueous Medium (The Interplay of Three Factors):

In water, three competing factors determine the overall stability of the ammonium cation:

- **Inductive Effect ($+\mathrm{I}$):** Stabilizes the cation in the order: $3^\circ > 2^\circ > 1^\circ$.
- **Solvation Effect (Hydrogen bonding with water):** Smaller cations with more $\mathrm{N-H}$ protons undergo greater hydration and stabilization:
$$\mathrm{R\overset{+}{N}H_3}\ (3\text{ H-bonds}) > \mathrm{R_2\overset{+}{N}H_2}\ (2\text{ H-bonds}) > \mathrm{R_3\overset{+}{N}H}\ (1\text{ H-bond})$$
- **Steric Hindrance:** Bulky alkyl groups crowd the nitrogen atom, hindering attack by protons and shielding the resulting cation from stabilizing hydration.

Definitive Orders for Aqueous Solutions (Memorize for NEET):

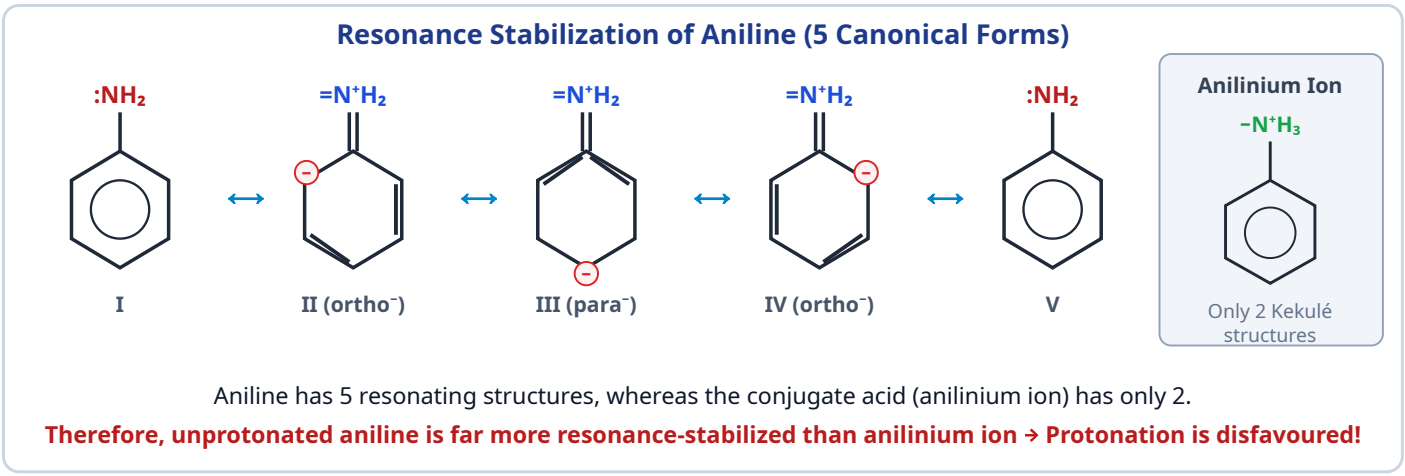
- **Ethyl-substituted series:**
$$\mathbf{(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3} \quad (2^\circ > 3^\circ > 1^\circ > \mathrm{NH_3})$$
- **Methyl-substituted series:**
$$\mathbf{(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3} \quad (2^\circ > 1^\circ > 3^\circ > \mathrm{NH_3})$$

Amine System	$\mathrm{p}K_b$ Value	Comparative Basicity Assessment
N-Ethylethanamine $[(\mathrm{C_2H_5})_2\mathrm{NH}]$	3.00	Strongest aliphatic base in aqueous ethyl series
N,N-Diethylethanamine $[(\mathrm{C_2H_5})_3\mathrm{N}]$	3.25	Intermediate in ethyl series
N-Methylmethanamine $[(\mathrm{CH_3})_2\mathrm{NH}]$	3.27	Strongest aliphatic base in aqueous methyl series
Ethanamine $[\mathrm{C_2H_5NH_2}]$	3.29	Primary aliphatic amine
Methanamine $[\mathrm{CH_3NH_2}]$	3.38	Primary aliphatic amine
N,N-Dimethylmethanamine $[(\mathrm{CH_3})_3\mathrm{N}]$	4.22	Weakest among simple methylamines (steric/solvation effects)
Ammonia $[\mathrm{NH_3}]$	4.75	Reference base lacking $+\mathrm{I}$ groups
Phenylmethanamine (Benzylamine)	4.70	Aliphatic amine (nitrogen not conjugated with ring)
N,N-Dimethylaniline	8.92	Aromatic tertiary amine
N-Methylaniline	9.30	Aromatic secondary amine
Benzenamine (Aniline)	9.38	Very weak base ($\mathrm{p}K_b$ elevated due to resonance)

Why is Aniline a Dramatically Weaker Base than Ammonia?

Aniline has a high $\mathrm{p}K_b$ of 9.38 (compared to 4.75 for $\mathrm{NH_3}$). In aniline, the lone pair on nitrogen is in direct conjugation with the π-electron system of the aromatic ring, making it delocalized into the ortho and para positions across five resonance contributors.

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Tutorial Diagram 2: Resonance structures of aniline vs. the anilinium ion demonstrating why aromatic amines are weak bases.

Substituent Effects on Benzenamine Basicity

- **Electron-Releasing Groups (ERG)** such as $-\mathrm{OCH_3}$ and $-\mathrm{CH_3}$ release electron density into the ring through inductive and resonance ($+R$) effects, stabilizing the conjugate acid and **increasing basic strength**.
- **Electron-Withdrawing Groups (EWG)** such as $-\mathrm{NO_2}$, $-\mathrm{CF_3}$, $-\mathrm{SO_3H}$, $-\mathrm{COOH}$, and $-\mathrm{X}$ withdraw electron density by $-I$ and $-R$ effects, destabilizing the conjugate acid and **decreasing basic strength**.

Comparative Basicity Order:

$$p\text{-Toluidine } (p\text{-}\mathrm{CH_3C_6H_4NH_2}) > \text{Aniline } (\mathrm{C_6H_5NH_2}) > p\text{-Nitroaniline } (p\text{-}\mathrm{NO_2C_6H_4NH_2})$$

Core Chemical Reactions

1. Acylation and Benzoylation

Primary and secondary amines react with acid chlorides, acid anhydrides, or esters via nucleophilic acyl substitution to form amides. The reaction is run in the presence of a base stronger than the amine (e.g., **pyridine**) to consume the liberated HCl and drive the equilibrium forward:

$$\mathrm{C_2H_5NH_2} + \mathrm{CH_3COCl} \xrightarrow{\text{Pyridine}} \mathrm{C_2H_5NHCOCH_3}\ (\text{N-Ethylethanamide}) + \mathrm{HCl}$$

$$\mathrm{C_6H_5NH_2} + (\mathrm{CH_3CO})_2\mathrm{O} \xrightarrow{\text{Pyridine}} \mathrm{C_6H_5NHCOCH_3}\ (\text{Acetanilide}) + \mathrm{CH_3COOH}$$

$$\mathrm{CH_3NH_2} + \mathrm{C_6H_5COCl} \xrightarrow{\text{NaOH/Pyridine}} \mathrm{CH_3NHCOC_6H_5}\ (\text{N-Methylbenzamide, Benzoylation}) +$$

2. Carbylamine Reaction (Isocyanide Test)

An essential diagnostic test: **Aliphatic and aromatic primary (1°) amines**, when warmed with chloroform ($\mathrm{CHCl_3}$) and ethanolic potassium hydroxide (KOH), yield an intensely foul-smelling **isocyanide (carbylamine)**:

$$\mathrm{R-NH_2} + \mathrm{CHCl_3} + 3\mathrm{KOH} \xrightarrow{\Delta} \mathrm{R-NC}\ (\text{Isocyanide / Carbylamine}) + 3\mathrm{KCl} + 3\mathrm{H_2O}$$

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Critical NEET Distinction: *Secondary (2°) and tertiary (3°) amines do not undergo the carbylamine reaction.* This reaction selectively identifies primary amino groups.

3. Reaction with Nitrous Acid ($\mathrm{HNO_2}$)

Nitrous acid is generated *in situ* by mixing sodium nitrite ($\mathrm{NaNO_2}$) with dilute mineral acid (HCl) at cold temperatures ($273\text{--}278\ \mathrm{K}$):

- **Primary Aliphatic Amines:** Form highly unstable alkyldiazonium salts that rapidly decompose, liberating nitrogen gas quantitatively and producing a mixture of alcohols, alkenes, and alkyl halides:

$$\mathrm{R-NH_2 + HNO_2 \xrightarrow{NaNO_2+HCl} \left[R-\overset{+}{N}_2\bar{Cl}\right] \xrightarrow{H_2O} ROH + \mathbf{N_2}\uparrow + HCl}$$

  Note: Quantitative evolution of $\mathrm{N_2}$ is utilized in analytical biochemistry to estimate amino acids and proteins.

- **Primary Aromatic Amines:** React at $273\text{--}278\ \mathrm{K}$ ($0\text{--}5^\circ\mathrm{C}$) to produce arenediazonium salts that are stabilized by resonance and persist in cold solution:

$$\mathrm{C_6H_5NH_2 + NaNO_2 + 2HCl \xrightarrow{273\text{--}278\ K} C_6H_5\overset{+}{N}_2\bar{Cl} + NaCl + 2H_2O}$$

4. Hinsberg's Test for Distinguishing $1^\circ, 2^\circ$, and 3° Amines

Benzenesulphonyl chloride ($\mathrm{C_6H_5SO_2Cl}$), known as **Hinsberg's reagent** (modern laboratories also use p-toluenesulphonyl chloride), differentiates primary, secondary, and tertiary amines based on the presence of replaceable hydrogen atoms on the nitrogen atom:

- **Primary Amines (1°):** React to form N-alkylbenzenesulphonamide. Because this product retains an acidic hydrogen on the sulfonamide nitrogen (acidified by the strongly electron-withdrawing $-\mathrm{SO_2}-$ group), it dissolves in aqueous KOH to form a clear solution. Subsequent acidification with mineral acid reprecipitates the insoluble sulfonamide:

$$\mathrm{C_6H_5SO_2Cl + RNH_2 \longrightarrow C_6H_5SO_2NHR + HCl \xrightarrow{KOH(aq)} Soluble\ potassium\ salt}$$

- **Secondary Amines (2°):** React to form N,N-dialkylbenzenesulphonamide. This product lacks an acidic hydrogen atom attached to nitrogen, making it completely **insoluble in aqueous** KOH:

$$\mathrm{C_6H_5SO_2Cl + R_2NH \longrightarrow C_6H_5SO_2NR_2\ (Insoluble\ in\ alkali) + HCl}$$

- **Tertiary Amines (3°):** Contain no replaceable hydrogen atom on the nitrogen atom and **do not react** with benzenesulphonyl chloride under these conditions.

Electrophilic Aromatic Substitution of Aniline

The $-\mathrm{NH_2}$ group donates electron density strongly through resonance ($+R$), making it a powerful *ortho/para*-directing and highly activating substituent toward electrophilic aromatic substitution.

(a) Bromination: Uncontrolled vs. Monosubstitution via Protection

- **Direct Bromination:** Aniline reacts immediately with bromine water at room temperature without requiring a Lewis acid catalyst, producing a white precipitate of **2,4,6-tribromoaniline**:

$$\mathrm{C_6H_5NH_2 + 3Br_2(aq) \longrightarrow 2,4,6\text{-}Tribromoaniline\downarrow\ (White\ ppt) + 3HBr}$$

- **Controlled Monobromination:** To prepare p-bromoaniline exclusively, the activating effect of the $-\mathrm{NH_2}$ group must be moderated. This is achieved by **acetylation** with acetic anhydride in pyridine to form acetanilide. The amide is then brominated with $\mathrm{Br_2}$ in glacial acetic acid, followed by acidic or alkaline hydrolysis to regenerate the amine:

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Mechanism of Protection: Delocalization of Nitrogen Lone Pair in Acetanilide



The nitrogen lone pair is delocalized onto the electronegative carbonyl oxygen via resonance.
Result: Reduced electron donation to the benzene ring → Ring activation drops → Clean monohalogenation at para!

Tutorial Diagram 3: Moderation of aniline's nucleophilic activating power via acetanilide resonance.

(b) Nitration: Anomalous Formation of m-Nitroaniline

Direct nitration of aniline with concentrated $\mathrm{HNO_3}$ and $\mathrm{H_2SO_4}$ at $288\,\mathrm{K}$ results in extensive oxidative degradation along with an unexpected product distribution:

$$\mathrm{C_6H_5NH_2} \xrightarrow[\mathrm{H_2SO_4, 288\,K}]{\mathrm{HNO_3}} \underset{(51\%)}{p\text{-nitroaniline}} + \underset{\mathbf{(47\%)}}{\mathbf{m\text{-nitroaniline}}} + \underset{(2\%)}{o\text{-nitroaniline}}$$

Why is 47% meta-isomer formed? In the strongly acidic nitrating mixture, aniline undergoes protonation to form the **anilinium ion ($-\overset{+}{N}H_3$)**. Because $-\overset{+}{N}H_3$ is a strong $-I$ group that deactivates the ring and directs substitution to the *meta* position, a substantial amount (47%) of m-nitroaniline is produced.

Synthetic remedy: Protect the amino group by acetylation with $(\mathrm{CH_3CO})_2\mathrm{O}$ in pyridine, nitrate the resulting acetanilide at $288\,\mathrm{K}$ (yielding p-nitroacetanilide exclusively), and subsequently hydrolyze to obtain pure p-nitroaniline.

(c) Sulphonation: Zwitterion Formation

Aniline reacts with concentrated sulfuric acid to yield anilinium hydrogensulfate, which upon prolonged heating at $453\text{--}473\,\mathrm{K}$ rearranges to form **sulphanilic acid** (p-aminobenzenesulfonic acid). Sulphanilic acid contains both an acidic sulfonic group and a basic amino group, existing predominantly as an internal dipolar **zwitterion**:

$$\mathrm{C_6H_5NH_2 + H_2SO_4 \longrightarrow C_6H_5\overset{+}{N}H_3HSO_4^- \xrightarrow{453\text{--}473\,K} H_2N - C_6H_4 - SO_3H \rightleftharpoons \mathbf{H_3\overset{+}{N} - C_6H_4 -}}$$

(d) Inability to Undergo Friedel–Crafts Reactions

NEET Question Classic: *Aniline does not undergo Friedel–Crafts alkylation or acylation reactions.*
The Lewis acid catalyst ($\mathrm{AlCl_3}$) forms an acid-base adduct with the basic lone pair of aniline:

$$\mathrm{C_6H_5\ddot{N}H_2 + AlCl_3 \longrightarrow C_6H_5NH_2^{\delta+} - AlCl_3^{\delta-}}$$

The resulting positive formal charge on nitrogen exerts a strong $-I$ effect, deactivating the benzene ring toward electrophilic attack by carbocations or acylium ions.

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7. Diazonium Salts: Preparation, Properties, and Synthetic Utility

Arenediazonium salts possess the general formula $\mathrm{Ar\overset{+}{N}_2\bar{X}}$, where Ar represents an aryl group and $\bar{\mathrm{X}}$ is an anion (e.g., $\mathrm{Cl^-}$, $\mathrm{Br^-}$, $\mathrm{HSO_4^-}$, $\mathrm{BF_4^-}$). They are named by appending "diazonium" to the parent arene, followed by the anion name (e.g., benzenediazonium chloride).

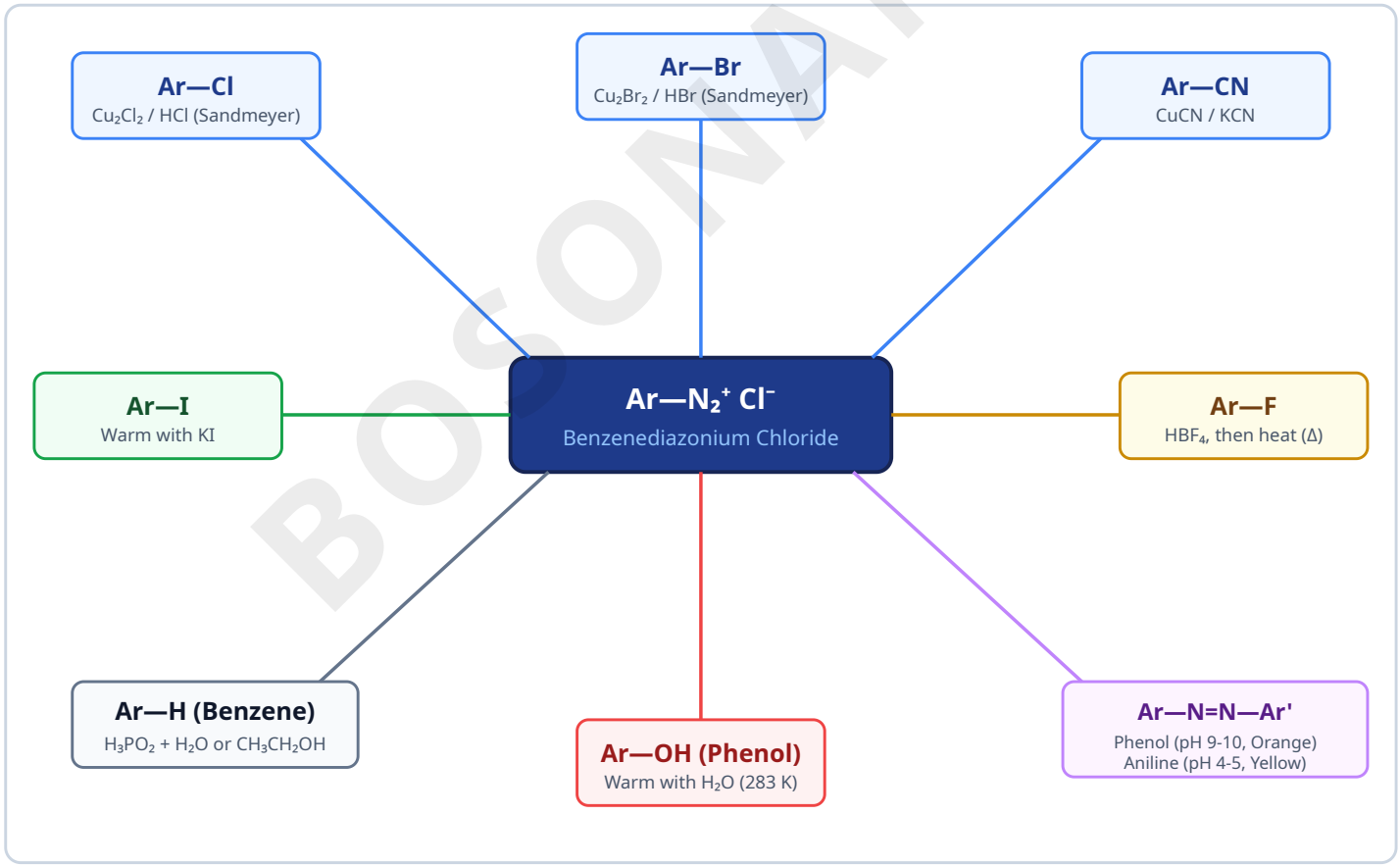
Diazotisation Reaction

Benzenediazonium chloride is prepared by treating cold aniline with nitrous acid (generated *in situ* from $\mathrm{NaNO_2 + HCl}$) at strictly maintained ice temperatures:

$$\mathrm{C_6H_5NH_2 + NaNO_2 + 2HCl \xrightarrow{273\text{--}278\ K\ (0\text{--}5^\circ C)} C_6H_5\overset{+}{N}_2\bar{Cl} + NaCl + 2H_2O}$$

- **Resonance Stability:** Arenediazonium ions are stabilized by resonance involving the aromatic ring, making them stable in solution at $273\text{--}278\ \mathrm{K}$. Alkyldiazonium ions lack this stabilization and decompose immediately with liberation of $\mathrm{N_2}$.
- **Physical Properties:** Benzenediazonium chloride is a colourless crystalline solid that is readily soluble in water. It is stable in cold aqueous solution, decomposes when warmed, and can detonate when dry. In contrast, benzenediazonium fluoroborate ($\mathrm{C_6H_5N_2^+BF_4^-}$) is water-insoluble and stable at room temperature.

Synthetic Reactions of Diazonium Salts



Tutorial Diagram 4: Roadmap of synthetic transformations utilizing arenediazonium salts.

Detailed Reaction Summary

1. **Sandmeyer Reaction:** Introducing $-\mathrm{Cl}$, $-\mathrm{Br}$, or $-\mathrm{CN}$ using cuprous salts ($\mathrm{Cu_2Cl_2/HCl}$, $\mathrm{Cu_2Br_2/HBr}$, $\mathrm{CuCN/KCN}$). Gives superior yields compared to the Gattermann reaction.
2. **Gattermann Reaction:** Introducing $-\mathrm{Cl}$ or $-\mathrm{Br}$ using copper powder and the corresponding halogen acid ($\mathrm{Cu/HCl}$ or $\mathrm{Cu/HBr}$).

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3. **Iodination:** Heating with aqueous potassium iodide (KI) yields iodobenzene without requiring copper catalysts:
$$\mathrm{ArN_2^+Cl^- + KI \longrightarrow Ar-I + KCl + N_2\uparrow}$$
4. **Fluorination (Balz–Schiemann Reaction):** Precipitation with fluoroboric acid ($\mathrm{HBF_4}$) produces arenediazonium fluoroborate, which upon heating yields the aryl fluoride:
$$\mathrm{ArN_2^+Cl^- + HBF_4 \longrightarrow ArN_2^+BF_4^- \downarrow \xrightarrow{\Delta} Ar-F + BF_3 + N_2\uparrow}$$
5. **Reductive Deamination (Replacement by $-\mathrm{H}$):** Diazonium salts are reduced to arenes by treatment with hypophosphorous acid ($\mathrm{H_3PO_2}$, phosphinic acid) or ethanol:
$$\mathrm{ArN_2^+Cl^- + H_3PO_2 + H_2O \longrightarrow Ar-H + N_2\uparrow + H_3PO_3 + HCl}$$
$$\mathrm{ArN_2^+Cl^- + CH_3CH_2OH \longrightarrow Ar-H + N_2\uparrow + CH_3CHO + HCl}$$
6. **Hydroxylation (Conversion to Phenols):** Warming the aqueous solution to $283\,\mathrm{K}$ causes hydrolysis to phenol:
$$\mathrm{ArN_2^+Cl^- + H_2O \xrightarrow{283\,K} Ar-OH + N_2\uparrow + HCl}$$
7. **Replacement by $-\mathrm{NO_2}$:** Heating diazonium fluoroborate with aqueous sodium nitrite in the presence of copper powder yields nitroarene:
$$\mathrm{ArN_2^+BF_4^- + NaNO_2 \xrightarrow{Cu,\Delta} Ar-NO_2 + N_2\uparrow + NaBF_4}$$
8. **Azo Coupling Reactions (Retention of the Diazo Group):** The arenediazonium ion serves as a weak electrophile, attacking activated aromatic rings (phenols and arylamines) at the *para* position to produce conjugated, intensely colored azo compounds:
   - **Coupling with Phenol:** Carried out in mildly basic medium ($\mathrm{pH} = 9\text{--}10$) to generate the phenoxide ion:
   $$\mathrm{C_6H_5N_2^+Cl^- + C_6H_5OH \xrightarrow{\bar{O}H,\ pH\ 9\text{--}10} \textbf{p-hydroxyazobenzene}\ (\text{Orange dye}) + Cl^- + H_2O}$$
   - **Coupling with Aniline:** Carried out in mildly acidic medium ($\mathrm{pH} = 4\text{--}5$):
   $$\mathrm{C_6H_5N_2^+Cl^- + C_6H_5NH_2 \xrightarrow{H^+,\ pH\ 4\text{--}5} \textbf{p-aminoazobenzene}\ (\text{Yellow dye}) + Cl^- + H_2O}$$

8. Multi-Step Synthesis Worked Examples

Worked Example 1: Synthesis of 2-Bromobenzoic Acid from 4-Nitrotoluene

Target: Convert 4-nitrotoluene into 2-bromobenzoic acid.

Step-by-step Synthetic Route:

1. **Regioselective Bromination:** Treat 4-nitrotoluene with $\mathrm{Br_2/Fe}$. The strongly activating $-\mathrm{CH_3}$ group directs bromination ortho to itself (meta to the deactivating $-\mathrm{NO_2}$ group), yielding 2-bromo-4-nitrotoluene.
2. **Reduction of the Nitro Group:** Reduce with $\mathrm{Sn/HCl}$ or $\mathrm{Fe/HCl}$ to obtain 3-bromo-4-methylaniline.
3. **Diazotisation:** Treat the resulting amine with $\mathrm{NaNO_2 + HCl}$ at $273\text{--}278\,\mathrm{K}$ to yield the corresponding arenediazonium salt.
4. **Reductive Deamination:** Remove the diazonium group using hypophosphorous acid ($\mathrm{H_3PO_2 + H_2O}$) to obtain 2-bromotoluene.
5. **Side-chain Oxidation:** Oxidize the methyl group using alkaline potassium permanganate ($\mathrm{KMnO_4/\bar{O}H}$) followed by acidic workup to produce 2-bromobenzoic acid.

BOSONANCE

Worked Example 2: Synthesis of 1,3,5-Tribromobenzene from Aniline

Target: Synthesize symmetrical 1,3,5-tribromobenzene from aniline.

Step-by-step Synthetic Route:

1. **Exhaustive Bromination:** Treat aniline with bromine water ($\mathrm{Br_2/H_2O}$) at room temperature to yield 2,4,6-tribromoaniline as a white precipitate.
2. **Diazotisation:** Treat 2,4,6-tribromoaniline with $\mathrm{NaNO_2}$ and HCl at $273\text{--}278\ \mathrm{K}$ to produce 2,4,6-tribromobenzenediazonium chloride.
3. **Reductive Removal of the Diazo Group:** React with hypophosphorous acid ($\mathrm{H_3PO_2}$) or boil with ethanol ($\mathrm{CH_3CH_2OH}$) to replace the diazonium group with hydrogen, cleanly yielding 1,3,5-tribromobenzene.

9. NEET-UG Practice Examination Questions

Original assessment items designed to reflect current NEET-UG examination patterns, including single-correct MCQs, assertion-reasoning, and statement evaluations.

Practice Question 1

Single Correct MCQ | Basicity Order

Which of the following represents the correct decreasing order of basic strength of the given amines in an **aqueous medium**?

(A) $(\mathrm{CH_3})_3\mathrm{N} > (\mathrm{CH_3})_2\mathrm{NH} > \mathrm{CH_3NH_2} > \mathrm{NH_3}$

(B) $(\mathrm{CH_3})_2\mathrm{NH} > \mathrm{CH_3NH_2} > (\mathrm{CH_3})_3\mathrm{N} > \mathrm{NH_3}$

(C) $(\mathrm{CH_3})_2\mathrm{NH} > (\mathrm{CH_3})_3\mathrm{N} > \mathrm{CH_3NH_2} > \mathrm{NH_3}$

(D) $\mathrm{CH_3NH_2} > (\mathrm{CH_3})_2\mathrm{NH} > (\mathrm{CH_3})_3\mathrm{N} > \mathrm{NH_3}$

Correct Answer: (B)

Worked Explanation:

In aqueous medium, the basic strength of methyl-substituted aliphatic amines is determined by an interplay of three factors: the inductive effect ($+I$), hydration/solvation of the conjugate ammonium cation, and steric hindrance. For methyl substituents, the secondary amine strikes the optimal balance between inductive donation and cation solvation, giving the order:

$$(\mathrm{CH_3})_2\mathrm{NH}\ (2^\circ) > \mathrm{CH_3NH_2}\ (1^\circ) > (\mathrm{CH_3})_3\mathrm{N}\ (3^\circ) > \mathrm{NH_3}$$

The respective $\mathrm{p}K_b$ values are $3.27 < 3.38 < 4.22 < 4.75$. A lower $\mathrm{p}K_b$ corresponds to a stronger base.

BOSONANCE

Practice Question 2

Single Correct MCQ | Name Reactions

An organic primary amine X was treated with chloroform and ethanolic potassium hydroxide to produce a foul-smelling liquid Y. Compound X can be prepared by the reduction of an amide with $\mathrm{Br_2}$ and NaOH. Identify the starting amide required to produce ethanamine as compound X:

(A) Ethanamide

(B) Propanamide

(C) Methanamide

(D) Butanamide

Correct Answer: (B)

Worked Explanation:

The reaction of compound X with $\mathrm{CHCl_3}$ and ethanolic KOH is the carbylamine test, confirming that X is a primary amine. The preparation of X from an amide using $\mathrm{Br_2}$ and NaOH is the Hoffmann bromamide degradation.
Because the Hoffmann bromamide degradation results in a *loss of one carbon atom*, preparing ethanamine (which has 2 carbons) requires a starting amide with $2 + 1 = 3$ carbons, namely **propanamide** ($\mathrm{CH_3CH_2CONH_2}$).

Practice Question 3

Assertion & Reason | Nitration of Aniline

Assertion (A): Direct nitration of aniline with a nitrating mixture produces a significant amount of m-nitroaniline (47%) in addition to the o- and p-isomers.
Reason (R): In strongly acidic media, aniline is protonated to form the anilinium ion, which acts as a meta-directing group.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Worked Explanation:

Under the strongly acidic conditions of nitration (concentrated $\mathrm{HNO_3} + \mathrm{H_2SO_4}$), the basic amino group of aniline undergoes protonation to yield the anilinium ion ($-\mathrm{N^+H_3}$). The positive charge exerts a strong electron-withdrawing $-I$ effect, deactivating the ring and directing incoming nitronium ions ($\mathrm{NO_2^+}$) to the *meta* position. This accounts for the significant 47% yield of m-nitroaniline.

BOSONANCE

Practice Question 4

Statement Evaluation | Gabriel Synthesis

Given below are two statements regarding Gabriel phthalimide synthesis:
Statement I: Gabriel phthalimide synthesis is widely employed for the convenient preparation of pure aromatic primary amines such as aniline.
Statement II: Aryl halides do not undergo nucleophilic substitution with the phthalimide anion under standard conditions.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct, but Statement II is incorrect.

(D) Statement I is incorrect, but Statement II is correct.

Correct Answer: (D)

Worked Explanation:

Statement I is incorrect because Gabriel phthalimide synthesis **cannot** be used to prepare aromatic primary amines; it is restricted to aliphatic primary amines.
Statement II is correct because aryl halides possess partial double-bond character in their $\mathrm{C-X}$ bond due to resonance delocalization, and the $\mathrm{sp^2}$-hybridized aromatic ring repels the nucleophilic phthalimide anion, preventing $\mathrm{S_N2}$ substitution.

Practice Question 5

Single Correct MCQ | Azo Coupling

Benzenediazonium chloride couples with phenol in a basic aqueous medium ($\mathrm{pH = 9\text{--}10}$) to produce an azo dye. What is the structure and characteristic color of this product?

(A) o -Hydroxyazobenzene, yellow dye

(B) p -Hydroxyazobenzene, orange dye

(C) p -Aminoazobenzene, yellow dye

(D) o -Aminoazobenzene, orange dye

Correct Answer: (B)

Worked Explanation:

In a mildly basic solution ($\mathrm{pH = 9\text{--}10}$), phenol is converted to the more nucleophilic phenoxide ion ($\mathrm{C_6H_5O^-}$). The benzenediazonium cation electrophilically attacks the sterically accessible and activated *para* position of the phenoxide ring to yield **p-hydroxyazobenzene**, which is an **orange dye**.

BOSONANCE

Practice Question 6

Single Correct MCQ | Hinsberg Test

An unknown amine with the molecular formula $C_4H_{11}N$ reacts with benzenesulphonyl chloride to form a white solid precipitate that is completely **insoluble** in aqueous potassium hydroxide. The original amine is:

(A) Butan-1-amine

(B) 2-Methylpropan-2-amine

(C) N-Ethylethanamine

(D) N,N-Dimethylethanamine

Correct Answer: (C)

Worked Explanation:

Reaction with benzenesulphonyl chloride followed by insolubility of the resulting sulfonamide in KOH is diagnostic for a **secondary (2°) amine**.

- Butan-1-amine and 2-methylpropan-2-amine are 1° amines; their sulfonamides retain an acidic hydrogen on nitrogen and dissolve in KOH.
- N,N-Dimethylethanamine is a 3° amine and does not react with benzenesulphonyl chloride.
- **N-Ethylethanamine (diethylamine)** is a 2° amine. It reacts to form N,N-diethylbenzenesulphonamide, which lacks an acidic hydrogen atom and is therefore insoluble in aqueous KOH.

Practice Question 7

Single Correct MCQ | Multi-Step Conversions

Consider the following multi-step reaction sequence:

$$CH_3CH_2Cl \xrightarrow{NaCN} \mathbf{A} \xrightarrow{LiAlH_4} \mathbf{B} \xrightarrow[273\text{--}278\ K]{HNO_2} \mathbf{C}$$

The final organic product $\mathbf{C}$ is:

(A) Propan-1-ol

(B) Ethanol

(C) Ethanoic acid

(D) Propanenitrile

Correct Answer: (A)

Worked Explanation:

1. $CH_3CH_2Cl + NaCN \longrightarrow CH_3CH_2CN$ (Propanenitrile, compound $\mathbf{A}$).
2. Reduction of propanenitrile with $LiAlH_4$ yields propan-1-amine:

$$CH_3CH_2CN \xrightarrow{LiAlH_4} CH_3CH_2CH_2NH_2 \quad (\text{Compound } \mathbf{B})$$

3. Primary aliphatic amines react with nitrous acid (HNO_2) to generate unstable diazonium salts, which decompose with liberation of N_2 gas to give **propan-1-ol** ($CH_3CH_2CH_2OH$, compound $\mathbf{C}$).

BOSONANCE

Practice Question 8

Single Correct MCQ | Industrial Reduction

In the reduction of nitrobenzene to aniline, the reaction using Fe scrap and HCl is preferred over Sn and HCl because:

(A) Iron acts as a stronger reducing agent than tin.

(B) The $\mathrm{FeCl_2}$ formed hydrolyses to regenerate HCl, requiring only a small amount of hydrochloric acid.

(C) The reaction with iron scrap avoids the formation of anilinium chloride.

(D) Tin forms an insoluble toxic complex with aniline that decreases the yield.

Correct Answer: (B)

Worked Explanation:

According to NCERT, reduction with iron scrap and hydrochloric acid is preferred industrially because the byproduct ferrous chloride ($\mathrm{FeCl_2}$) undergoes hydrolysis in the aqueous reaction mixture:

$$\mathrm{FeCl_2 + 2H_2O \longrightarrow Fe(OH)_2 + 2HCl}$$

This regenerates hydrochloric acid, meaning only a catalytic or small initiating quantity of HCl is required to drive the reduction to completion.

10. Common Student Pitfalls and High-Yield Revision Checklist

Frequent Traps in NEET Questions on Nitrogen Compounds

- **Confusing Aqueous and Gas-Phase Basicity Orders:** Remember that in the gas phase, basicity follows the pure inductive effect: $3^\circ > 2^\circ > 1^\circ > \mathrm{NH_3}$. In aqueous solution, the order changes to $2^\circ > 3^\circ > 1^\circ > \mathrm{NH_3}$ for ethyl groups (231) and $2^\circ > 1^\circ > 3^\circ > \mathrm{NH_3}$ for methyl groups (213).
- **Overlooking Carbon Count in the Hoffmann Bromamide Reaction:** Hoffmann bromamide degradation produces an amine with **one fewer carbon** than the starting amide. By contrast, reduction of nitriles or amides using $\mathrm{LiAlH_4}$ preserves the original carbon chain length.
- **Misidentifying Aromatic Gabriel Syntheses:** Gabriel phthalimide synthesis *cannot* yield aniline because bromobenzene will not undergo nucleophilic substitution with the phthalimide anion.
- **Assuming Aniline Readily Undergoes Friedel–Crafts Reactions:** Aniline fails to undergo Friedel–Crafts alkylation or acylation because $\mathrm{AlCl_3}$ coordinates to the lone pair of nitrogen, generating a deactivated ring.
- **Missing the Meta Isomer in Direct Nitration:** Direct nitration of aniline produces 51% para, 47% meta, and only 2% ortho isomer. Do not assume nitration produces only ortho and para derivatives.

BOSONANCE

Quick NEET Revision Checklist

- ✔ Amines feature an sp^3-hybridized nitrogen with a pyramidal molecular geometry; the $\mathrm{C-N-C}$ bond angle in $(\mathrm{CH_3})_3\mathrm{N}$ is compressed to 108°.
- ✔ Biological amine roles: Ephedrine/adrenaline are 2° amines; benadryl is a 3° amine; novocain is a synthetic local anaesthetic.
- ✔ Hofmann bromamide degradation consumes 1 mole of $\mathrm{Br_2}$ and 4 moles of NaOH per mole of primary amide.
- ✔ Carbylamine test is specific to 1° aliphatic and 1° aromatic amines; secondary and tertiary amines give no reaction.
- ✔ Hinsberg's reagent ($\mathrm{C_6H_5SO_2Cl}$): 1° amine product dissolves in KOH; 2° amine product does not dissolve in KOH; 3° amine does not react.
- ✔ Arenediazonium salts undergo Sandmeyer reactions with $\mathrm{Cu(I)}$ halides to give ArCl, ArBr, and ArCN; with KI to give ArI; and with $\mathrm{HBF_4}$ followed by heat to give ArF (Balz–Schiemann).
- ✔ Reduction of diazonium salts with $\mathrm{H_3PO_2}$ or ethanol replaces the diazonium group with $-\mathrm{H}$, producing the parent arene.
- ✔ Azo coupling: Phenol couples in mildly basic conditions ($\mathrm{pH} = 9\text{--}10$) to yield an orange dye; aniline couples in mildly acidic conditions ($\mathrm{pH} = 4\text{--}5$) to yield a yellow dye.

11. Chapter Summary

Amines are alkyl and aryl derivatives of ammonia classified into primary, secondary, and tertiary categories. They possess a pyramidal geometry about an sp^3-hybridized nitrogen atom. Primary and secondary amines display elevated boiling points due to intermolecular hydrogen bonding ($1^\circ > 2^\circ > 3^\circ$).

The basicity of amines is governed by a balance of inductive effects, steric factors, and cation hydration in water, making secondary amines the strongest bases among simple aliphatic systems. Aromatic amines are significantly weaker bases than ammonia because the nitrogen lone pair is delocalized into the π-system of the ring. Primary amines are distinguished through the carbylamine reaction and Hinsberg's test.

Electrophilic substitution of aniline is strongly activated toward the *ortho* and *para* positions; controlling substitution requires temporary protection of the amino group via acetylation to acetanilide. Arenediazonium salts, synthesized by cold diazotisation of primary aromatic amines, serve as versatile synthetic intermediates for preparing functionalized arenes, including aryl halides, cyanides, phenols, and azo dyes.

BOSONANCE